Singapore School Kinderland

Secondary 1 (Sec 1) Computer Science — Term 2 Comprehensive Practice Worksheet

Topics: Error Detection Methods (TB Pg 54–62) & Encryption (TB Pg 63–74) | Total Marks: 40

Student Name: _____ **Class:** Sec 1 | **Date:** _____ **Marks:** / 40

SECTION A: PARITY CHECKS & PARITY BLOCKS (10 MARKS)

Q1. Parity Bit Calculation *[4 Marks]*

A computer system transmits 7-bit ASCII data. Add the required Parity Bit (most significant bit) for each byte below:

Byte	7-bit Data Payload	Parity System	Required Parity Bit (0 or 1)	Complete 8-bit Transmitted Byte
A	1 1 0 0 1 0 1	Even Parity		
B	1 0 1 1 1 1 0	Even Parity		
C	0 1 1 0 0 0 1	Odd Parity		
D	1 1 1 0 1 1 1	Odd Parity		

Q2. Parity Block Matrix Analysis & Auto-Correction *[6 Marks]*

A block of four bytes was transmitted using **Even Parity**. An extra parity byte was appended to the bottom:

Row	Bit 7 (Parity)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Row Status (OK / Error)
Byte 1	1	0	1	1	0	0	1	0	
Byte 2	0	1	1	0	1	0	1	1	
Byte 3	1	1	0	1	1	0	0	1	
Byte 4	1	0	0	1	0	1	1	0	
Parity Byte	**1**	**0**	**0**	**1**	**0**	**1**	**1**	**0**	

(a) Identify the corrupted row and column: Corrupted Row: _____ | Corrupted Column: _____ *[2]*

(b) State the original corrupted bit value: _____ and state the corrected bit value: _____ *[2]*

(c) Explain why this parity block cannot correct an error if two bits flip in the same row and column: *[2]*

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